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1. INTRODUCTION

1.1 Project Overview

The House Price Analysis project aims to analyze housing market data using Tableau and present meaningful insights through interactive dashboards and visualizations. The project examines various housing attributes such as house prices, house age, renovation status, bedrooms, bathrooms, floors, and sales trends to help users better understand the housing market. Interactive dashboards enable users to explore the data efficiently and identify important patterns that support informed decision-making.

1.2 Purpose

The primary purpose of this project is to transform raw housing data into meaningful visual information using Tableau. The project helps home buyers, property sellers, and real estate analysts understand housing market trends, compare property characteristics, and identify the major factors influencing house prices.

2. IDEATION PHASE

2.1 Problem Statement:

Customer Problem Statement :

The **House Price Analysis** project is designed to understand the challenges faced by home buyers, property sellers, and real estate professionals when analyzing housing market data. This project focuses on identifying the difficulties users encounter while comparing house prices, property features, renovation status, and market trends. By clearly defining these problems, the team aims to develop an effective data visualization solution that simplifies complex housing information and supports better decision-making.

A well-defined customer problem statement helps the team understand user needs and expectations from different perspectives. It enables the identification of important factors affecting house prices and transforms raw housing data into meaningful visual insights. Using **Tableau**, the project presents interactive dashboards and stories that allow users to explore housing trends, compare property characteristics, and make informed decisions based on accurate and easy-to-understand visual analysis.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	I am a home buyer.	Find a house that fits my budget and needs.	I cannot easily compare house prices, locations, and property features.	Housing data is scattered and difficult to understand.	Confused and unsure about making the right buying decision.
PS-2	I am a real estate agent or property analyst.	Analyze housing prices and market trends.	It is difficult to identify price patterns and important factors affecting house values	Large datasets are complex and require proper visualization and analysis.	Time-consuming and less confident in providing accurate recommendations.

			using raw data.		
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2.2 Empathy Map Canvas:

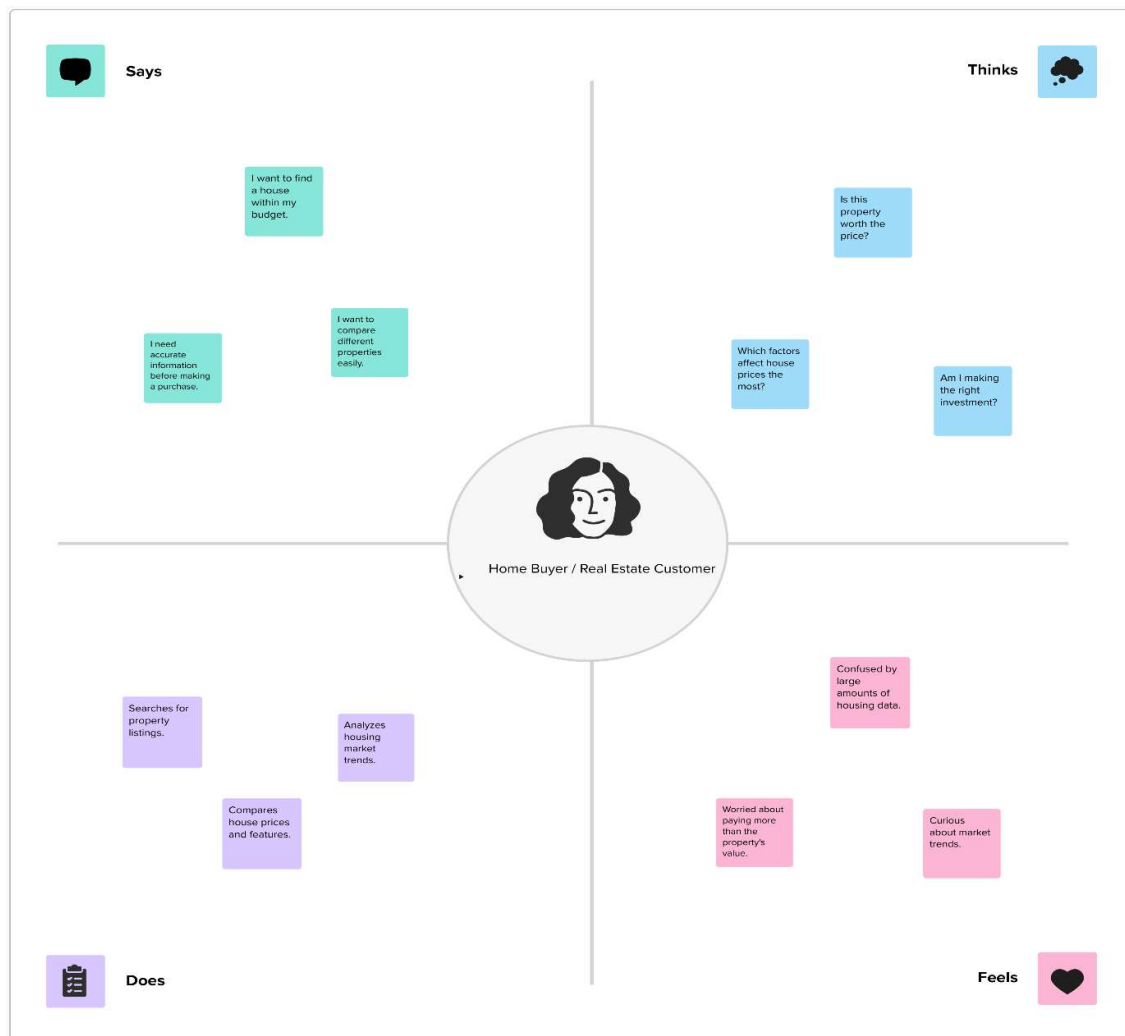
Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to help teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau:



2.3 Brainstorming

Brainstorm & Idea Prioritization :

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Brainstorm & idea prioritization

This brainstorming session is conducted to generate, discuss, and prioritize ideas for the House Price Analysis project. The team identifies the best approaches for analyzing housing data, discovering market trends, and designing interactive Tableau dashboards that provide clear and meaningful insights.

Before you collaborate

Before starting the project, the team discussed the challenges in understanding housing market trends and identifying the factors affecting house prices. Team members reviewed the dataset, defined project objectives, assigned responsibilities, and planned the workflow for analyzing and visualizing housing data using Tableau.

- Team gathering**
Form a project team and discuss the objectives, project scope, and expected outcomes.
- Set the goal**
Develop an interactive Tableau dashboard that provides meaningful insights into house prices, property features, and market trends.
- Learn how to use the facilitation tools**
Understand Tableau, data preprocessing techniques, and visualization methods before beginning the project.

[Open article](#)

Define your problem statement

How might we analyze and visualize housing market data to help users understand house price trends and make better property-related decisions?

Key rules of brainstorming

To run a smooth and productive session:

- Stay in topic
- Encourage wild ideas
- Defer judgment
- Listen to others
- Go for volume
- If possible, be visual

Step-2: Brainstorm, Idea Listing and Grouping

Brainstorm

Write down ideas that help analyze housing market data, identify factors affecting house prices, and create meaningful Tableau visualizations. Discuss all ideas with the team and select the most effective solution for the project.

- Collect housing dataset
- Identify required attributes
- Define project objective
- Remove missing values
- Clean duplicate records
- Standardize data
- Analyze house prices
- Complete house age
- Study renovation status
- Analyze bedrooms
- Analyze bathrooms
- Analyze floors
- Identify market trends
- Compare property features
- Analyze price distribution
- Create bar charts
- Create line charts
- Create pie charts
- Build interactive dashboard
- Create Tableau story
- Apply dashboard
- Generate business insights
- Support decision making
- Prepare project report

Group ideas

Group similar ideas together based on their purpose and importance. Organize the brainstormed ideas into meaningful categories to identify the most suitable approach for developing the House Price Analysis dashboard.

Step-3: Idea Prioritization

3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Import housing dataset (CSV) into Tableau
		Verify dataset integrity and structure
FR-2	Data Preprocessing	Remove duplicate records
		Handle missing values and clean the dataset
FR-3	Data Visualization	Create charts and graphs for housing analysis
		Build interactive dashboards and Tableau Story
FR-4	Data Analysis	Analyze house prices and property features
		Generate business insights and trend analysis

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboards should be simple, interactive, and easy to understand.
NFR-2	Performance	Visualizations should load quickly and respond efficiently.
NFR-3	Reliability	The analysis should produce consistent and accurate results.
NFR-4	Data Accuracy	Housing data should be processed correctly without loss of information.
NFR-5	Scalability	The dashboard should support larger housing datasets in the future.
NFR-6	Maintainability	Dashboards should be easy to update when new housing data is available.

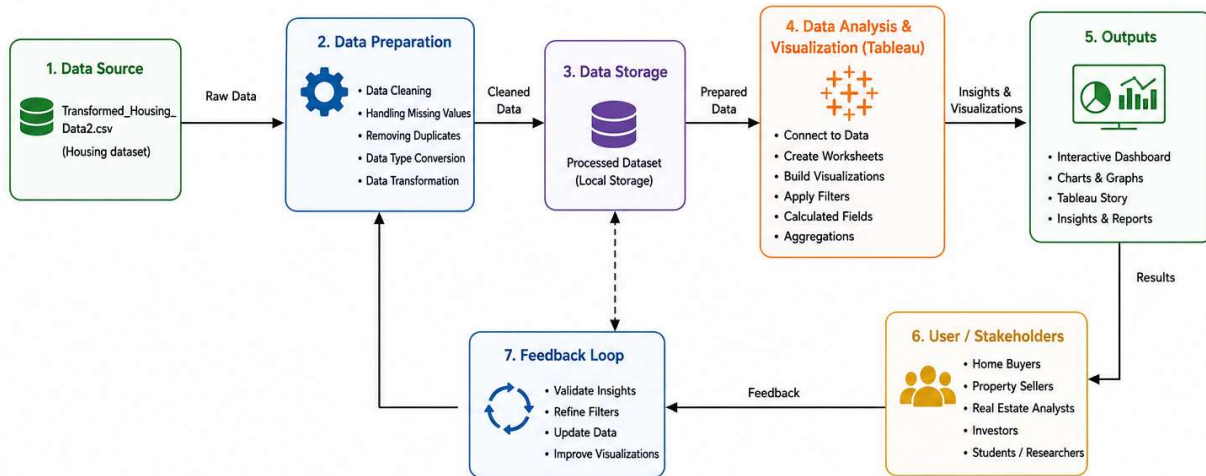
3.3 Data Flow Diagram

Data Flow Diagrams:

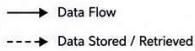
A Data Flow Diagram (DFD) illustrates how housing data moves through the House Price Analysis system. It represents the flow of data from the input dataset through preprocessing and analysis to the Tableau dashboards, where meaningful visualizations and business insights are generated. The DFD helps in understanding how data is processed and transformed into valuable information for users.

DATA FLOW DIAGRAM

House Price Analysis Using Tableau



Legend



Flow Description

- 1. Data Source:** Raw housing data is collected from CSV file (Transformed_Housing_Data2.csv).
- 2. Data Preparation:** The raw data is cleaned, processed and transformed for analysis.
- 3. Data Storage:** The processed dataset is stored locally and used for analysis.
- 4. Data Analysis & Visualization:** Using Tableau, data is connected, analyzed and visualized with interactive filters and calculations.
- 5. Outputs:** Dashboards, charts, graphs and stories are created to derive insights and reports.
- 6. Users / Stakeholders:** Insights are used by stakeholders to make informed decisions.
- 7. Feedback Loop:** Feedback from users helps refine data, filters and visualizations for continuous improvement.

User Stories

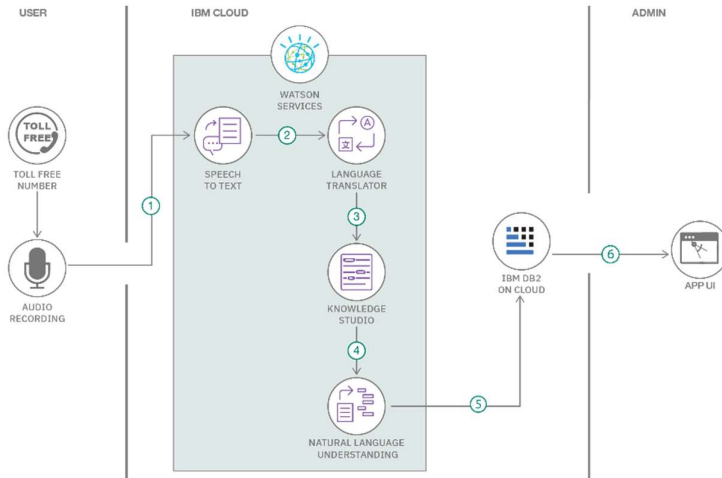
Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
User	Dataset Import	USN-1	As a user, I can import the housing dataset into Tableau.	Dataset is loaded successfully.	High	Sprint-1
User	Data Cleaning	USN-2	As a user, I can clean and preprocess the housing data before analysis.	Clean dataset is ready for visualization.	High	Sprint-1
User	Dashboard	USN-3	As a user, I can view interactive dashboards showing housing market trends.	Dashboards display accurate visualizations.	High	Sprint-1
User	Filters	USN-4	As a user, I can apply filters to analyze specific housing information.	Filtered results are displayed correctly.	Medium	Sprint-2
User	Analysis	USN-5	As a user, I can compare house prices based on different property features.	Comparative analysis is available.	High	Sprint-2
User	Reports	USN-6	As a user, I can identify market trends and business insights from the dashboards.	Insights are presented clearly.	Medium	Sprint-2

3.4 Technology Stack:

Technical Architecture:

The proposed **House Price Analysis** system is designed to analyze housing market data and present meaningful insights through interactive Tableau dashboards. The architecture consists of data collection, preprocessing, data analysis, visualization, and dashboard presentation. The processed housing data is transformed into interactive charts, dashboards, and stories that help users understand house price trends and property characteristics.



Guidelines:

- Import housing dataset (CSV).
- Perform data cleaning and preprocessing.
- Analyze housing market data.
- Create interactive visualizations.
- Build Tableau dashboards and story.
- Generate business

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	Data Source	Housing dataset in CSV format	CSV File
2	Data Preprocessing	Data cleaning and transformation	Tableau Prep / Tableau Desktop
3	Data Analysis	Analyze housing data and identify trends	Tableau Desktop
4	Data Visualization	Create interactive charts and dashboards	Tableau
5	Dashboard	Present business insights through dashboards	Tableau Dashboard
6	Story	Display analysis using Tableau Story	Tableau Story
7	File Storage	Store datasets and Tableau workbook	Local File System
8	Operating System	Platform used for development	Windows 10/11
9	Development Tool	Dashboard development environment	Tableau Desktop
10	Output	Interactive reports and visualizations	Tableau Public / Tableau Desktop

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Visualization Tool	Interactive dashboard development	Tableau Desktop
2	Data Accuracy	Accurate preprocessing and visualization	Tableau
3	Scalability	Supports additional housing data in future	Tableau
4	Availability	Dashboard can be accessed locally or through Tableau Public	Tableau Public
5	Performance	Fast filtering, visualization, and dashboard interaction	Tableau

4. PROJECT DESIGN

4.1 Problem Solution Fit:

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.

Define CS: fit into CC	1. CUSTOMER SEGMENT(S) CS Who is your customer? i.e. who will use this solution? • Home Buyers • Property Sellers • Real Estate Analysts • Investors • Students / Researchers	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices of solutions? i.e. limited time, lack of tools, cost, data complexity, technical skills. • Difficulty in understanding large raw data • Lack of tools to analyze housing trends • Limited time for manual analysis • High cost of advanced analytics tools • Need for interactive and easy-to-use solution	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. Excel sheets, reports, real estate websites, manual analysis. • Excel spreadsheets • Real estate websites • Manual reports • Data analysis consulting services • Basic data visualization tools	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which job(s)-to-be-done (or problems) do you address for your customers? There could be more than one, explore different sides. • Need to understand housing market trends • Need to compare property features • Need to analyze renovation impact on prices • Need to identify factors affecting house prices • Need to visualize data for better decisions	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? i.e. lack of accessible insights, scattered data, complex data analysis. • Housing data is complex and unstructured • Information is scattered across multiple sources • Manual analysis is time-consuming and error-prone • Lack of interactive dashboards for exploration • Difficult to identify important patterns & trends	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? i.e. directly related, find the right solution, calculate usage and benefits; indirectly associated. • Search for property information online • Compare prices and features manually • Ask real estate agents • Use Excel to analyze data • Make decisions based on limited data	
Identify through TR & EM	3. TRIGGERS TR What triggers customers to act? i.e. seeing property prices rise, planning to buy/sell, news, investment opportunities. • Increase in property prices • Planning to buy or sell a property • Market trends and news • Investment opportunities • Need for data-driven decisions	10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution first. If in the canvas, and check how much it fits really. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour. • Interactive Tableau dashboards for housing analysis • Clean and processed dataset for accurate insights • Visualizations: charts, graphs, KPIs, and trends • Filters for dynamic data exploration • Story to communicate insights effectively • Easy-to-use and user-friendly interface	8. CHANNELS OF BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels. • Tableau dashboards (local / Tableau Public) • Website / Portfolio • Email reports • Online presentations 8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels. • Meetings and discussions • Printed reports • Offline presentations	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? i.e. confused, uncertain, stressed → relieved, satisfied, confident. Before: Confused, uncertain, overwhelmed, time-consuming After: Satisfied, informed, confident, relieved			

4.2 Proposed Solution

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Understanding housing market trends and identifying the factors that influence house prices is difficult when dealing with large and complex datasets. Users require an interactive and easy-to-understand visualization system to analyze housing information effectively.
2	Idea / Solution Description	Develop an interactive House Price Analysis dashboard using Tableau. The solution analyzes housing data, visualizes market trends, compares property characteristics, and presents meaningful insights through dashboards, charts, and stories to support informed decision-making.
3	Novelty / Uniqueness	The project combines multiple housing attributes such as house price, age, renovation status, bedrooms, bathrooms, floors, and sales trends into a single interactive Tableau dashboard, allowing users to perform detailed analysis using filters and visualizations.
4	Social Impact / Customer Satisfaction	The project helps home buyers, property sellers, and real estate analysts understand housing market trends more easily. Interactive visualizations simplify complex data, enabling users to make informed and confident property-related decisions.
5	Business Model (Revenue Model)	The dashboard can be used by real estate agencies, property consultants, and market analysts for housing market analysis. Organizations may use the dashboard as a decision-support tool or integrate it into their internal reporting systems.
6	Scalability of the Solution	The solution can be extended by incorporating larger housing datasets, additional property attributes, regional comparisons, and future housing market data to provide more comprehensive and detailed analysis.

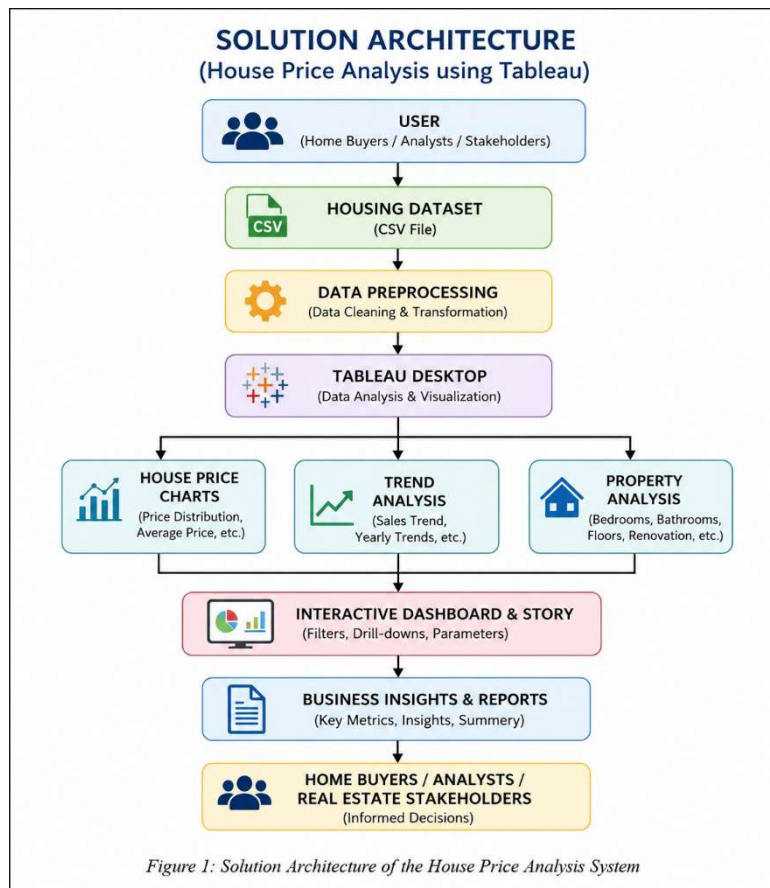
4.3 Solution Architecture

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture of the House Price Analysis System:



5. PROJECT PLANNING & SCHEDULING:

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority
Sprint-1	Data Collection	USN-1	Collect the housing dataset and verify its quality.	2	High
Sprint-1	Data Preprocessing	USN-2	Clean the dataset by removing duplicates and handling missing values.	3	High
Sprint-1	Data Transformation	USN-3	Organize and transform data for analysis in Tableau.	2	High
Sprint-2	Data Analysis	USN-4	Analyze house prices, house age, renovation status, bedrooms, bathrooms, and floors.	3	High
Sprint-2	Dashboard Design	USN-5	Create interactive charts and dashboards in Tableau.	5	High
Sprint-3	Story Creation	USN-6	Develop a Tableau Story to present business insights.	3	Medium
Sprint-3	Testing	USN-7	Validate dashboard accuracy and check visualizations.	2	Medium
Sprint-4	Documentation	USN-8	Prepare the final report and presentation.	2	High
Sprint-4	Deployment	USN-9	Publish the Tableau dashboard and complete project submission.	2	Medium
Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	7	5 Days	01 Mar 2025	05 Mar 2025	7	05 Mar 2025
Sprint-2	8	5 Days	06 Mar 2025	10 Mar 2025	8	10 Mar 2025
Sprint-3	5	5 Days	11 Mar 2025	15 Mar 2025	5	15 Mar 2025
Sprint-4	4	5 Days	16 Mar 2025	20 Mar 2025	4	20 Mar 2025

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing:

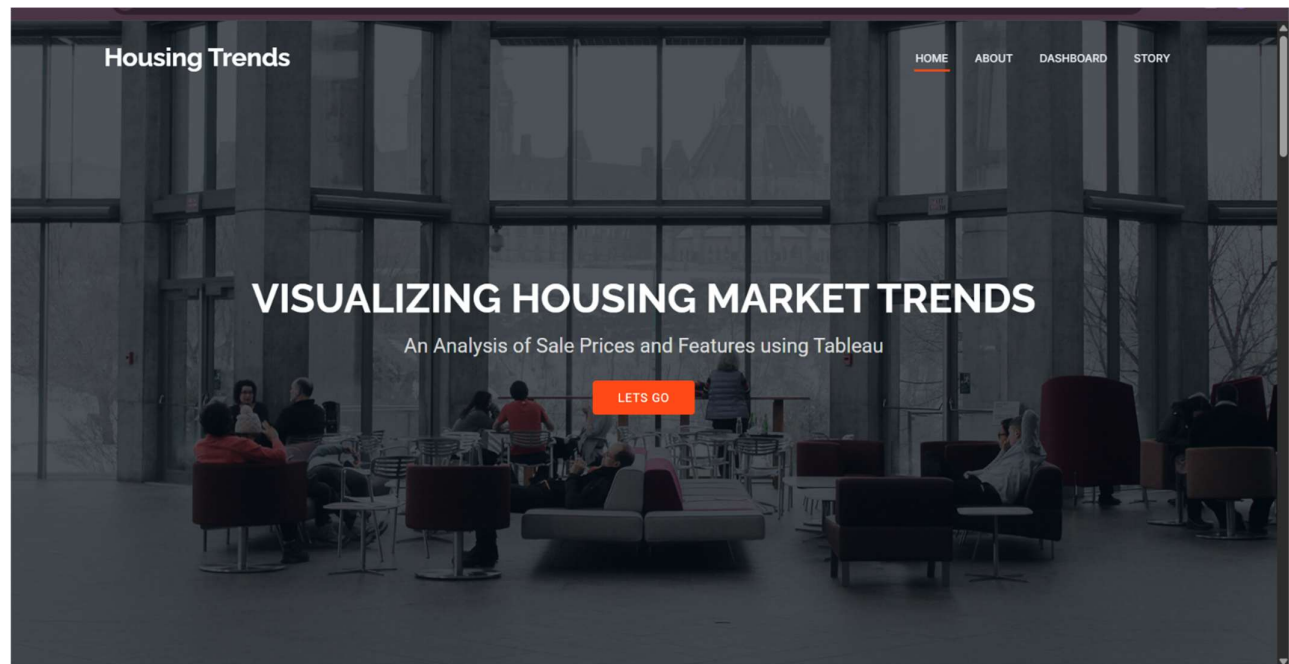
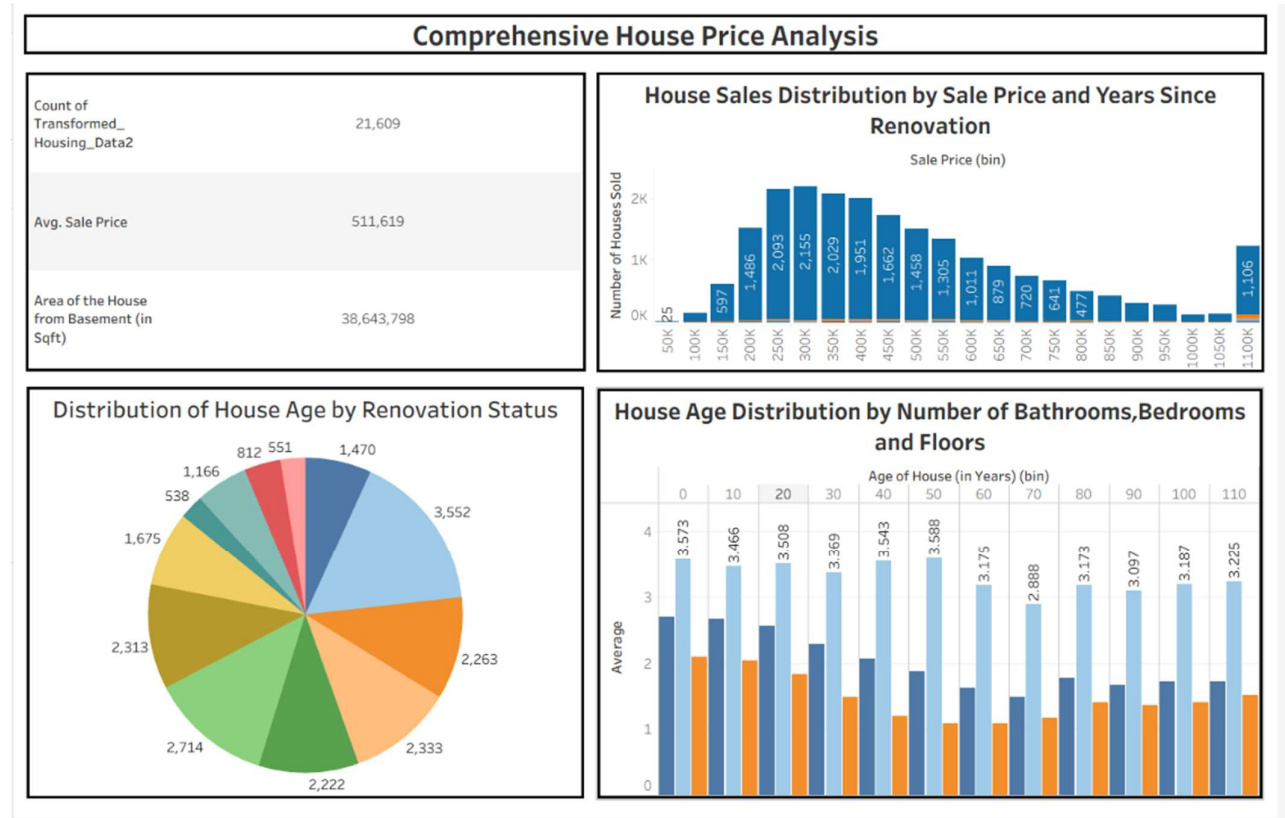
Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values
1. 1	Data Rendered	Housing dataset Transformed_Housing_Data2.csv was successfully imported into Tableau. All records and required fields were loaded correctly without data corruption.
2. 2	Data Preprocessing	Missing values were handled, duplicate records were removed (where applicable), data types were verified, and the dataset was prepared for visualization in Tableau.
3	Utilization of Filters	Interactive filters were used to analyze housing data based on attributes such as House Age, Renovation Status, Bedrooms, Bathrooms, Floors, and Years Since Renovation , enabling dynamic exploration of the dataset.
4	Calculated Fields Used	Basic Tableau calculations and aggregations were used to summarize housing data for visualization. Calculated measures supported analysis of house price distribution and sales trends. <i>(If you did not create any custom calculated fields, you can write: No custom calculated fields were created.)</i>
5	Dashboard Design	4 visualizations were created to analyze different aspects of the housing dataset, including house age distribution, renovation status, property features, and total sales trends.
6	Story Design	1 Tableau Story was created to present the overall analysis by combining dashboard visualizations and explaining key insights related to house prices and housing market trends.

7. RESULTS

7.1 Output Screenshots



127.0.0.1:5000/#about

Housing Trends

HOMEABOUTDASHBOARDSTORY



Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

- ✓ The goal is to address challenges in understanding the factors that influence house prices and sales trends.
- ✓ By analysing comprehensive housing data, including the total sales by years since renovation, house age distribution by the number of bathrooms, bedrooms, and floors, and the impact of renovations on house age, the company aims to uncover key insights.
- ✓ Utilizing Tableau for this analysis, the objective is to visualize and interpret patterns in the housing market to inform strategic decisions, optimize pricing strategies, and enhance overall market competitiveness.

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DASHBOARD

Housing Trends

HOMEABOUTDASHBOARDSTORY

Comprehensive House Price Analysis

Count of Transformed...
Housing_Data2

21,609

Avg. Sale Price

511,619

Area of the House from
Basement (in Sqft)

38,643,798

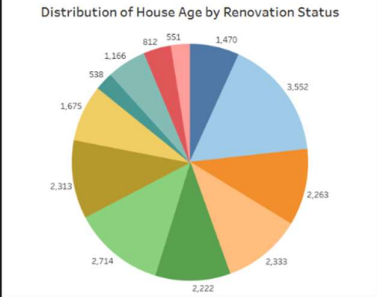
House Sales Distribution by Sale Price and Years Since Renovation

Sale Price (bin)



Sale Price (bin)	Number of Houses Sold
50K	15
100K	397
150K	1446
200K	2093
250K	2185
300K	2028
350K	1865
400K	1622
450K	1458
500K	1395
550K	1031
600K	879
650K	720
700K	601
750K	477
800K	405
850K	308
900K	208
950K	108
1000K	108
1050K	108
1100K	108
1150K	108
1200K	108
1250K	108
1300K	108
1350K	108
1400K	108
1450K	108
1500K	108

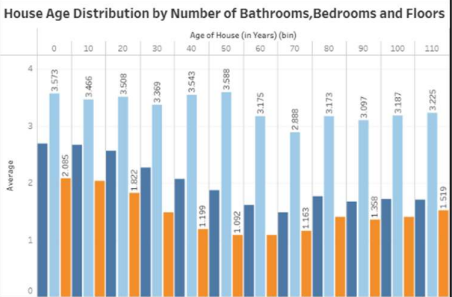
Distribution of House Age by Renovation Status



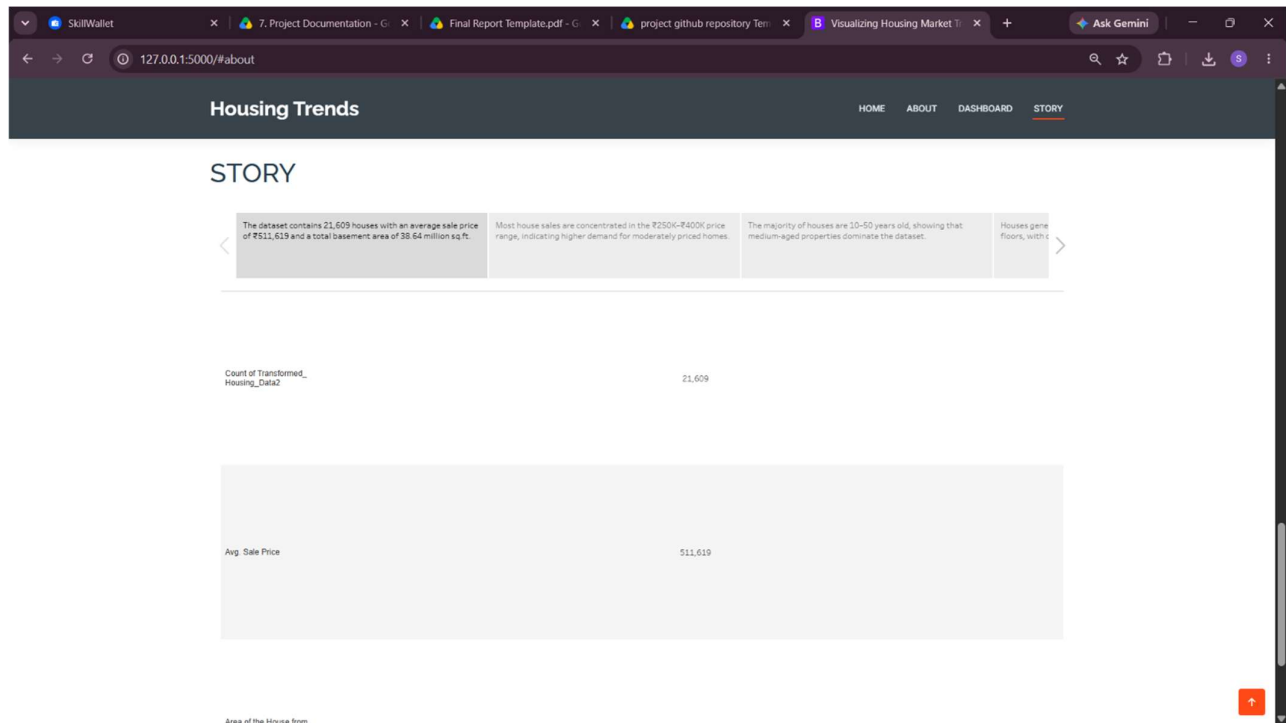
Renovation Status	Count
1	1,166
2	812
3	551
4	1,470
5	3,552
6	2,263
7	2,333
8	2,222
9	2,714
10	2,313
11	1,675
12	538

House Age Distribution by Number of Bathrooms, Bedrooms and Floors

Age of House (in Years) (bin)



Age of House (in Years) (bin)	Average
0	3.573
10	2.895
20	3.466
30	3.508
40	1.829
50	3.369
60	3.543
70	3.588
80	3.175
90	2.888
100	3.173
110	3.097



8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive visualization of housing data.
- Easy comparison of property features.
- Better understanding of market trends.
- Supports informed decision-making.
- User-friendly Tableau dashboards.

Disadvantages

- Results depend on dataset quality.
- Limited to available housing data.
- Requires Tableau software for editing dashboards.
- Dashboard performance may decrease with very large datasets.

9. CONCLUSION

The House Price Analysis project successfully demonstrates how Tableau can transform raw housing data into meaningful visual insights. Through interactive dashboards, charts, and stories, the project enables users to analyze housing market trends, compare property characteristics, and understand factors influencing house prices. The developed solution provides an effective platform for exploring housing data and supports better data-driven decision-making.

10. FUTURE SCOPE

- Include additional housing datasets.
- Add regional and city-wise comparisons.
- Incorporate real-time housing market updates.
- Develop predictive price trend analysis.
- Enhance dashboard interactivity with more filters and visualizations.

11. APPENDIX:

Dataset Link

<https://www.kaggle.com/datasets/rituparnaghosh18/transformed-housing-data-2>

Tableau Workbook

https://drive.google.com/file/d/1pGKxqme9wOUixaN9BYYfPCzX_YWUH4q-/view?usp=sharing

Tableau Public Link

https://public.tableau.com/app/profile/shaik.nizaamuddin/viz/Book1_17826726804080/Dashboard1

https://public.tableau.com/app/profile/shaik.nizaamuddin/viz/Book2_17826729225890/Story1

GitHub Repository:

<https://github.com/nizaam-07/Visualizing-Housing-Market-Trends-An-Analysis-of-Sale-Prices-and-Features-using-Tableau.git>